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High Density Lipoprotein-3 Heterogeneity in Subjects with the Apo-AI_{Milano} Variant*

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The structure of high density lipoproteins (HDL) isolated from subjects with the AI_{Milano} (AI_M) apoprotein variant was studied by the use of the cross-linking reagent dimethylsuberimidate. The HDL_2 subfraction is markedly reduced, as compared to control subjects; the HDL_3 subfraction, on the other hand, shows a marked heterogeneity, being characterized by at least three particle subpopulations, identified as HDL_3-I, HDL_3-II, and HDL_3-III. The HDL_3 fraction purified from the AI_{Milano} subjects eluted as a symmetrical peak from a 6% agarose column, corresponding to a unimodal particle size distribution. The content of the different HDL_3 particles, detected by cross-linking, varied widely along the elution profile, the tail of the peak being enriched in anomalous particles of very small size (HDL_3-III). Apoprotein compositional studies indicated that these small HDL_3-III may be enriched in the AI_M monomer, the larger particles containing more AI_M-AII complexes and AI_M dimers. All the anomalous HDL_3 particles are triglyceride enriched, with a decreased cholesterol ester content. They may be an intermediate product in the cholesterol transfer chain between HDL and very low density lipoproteins, or be generated during interconversion of HDL. These particles may have a functional role in tissue cholesterol homeostasis; their unusual compositional changes may help explain the protection of the studied subjects from tissue cholesterol deposition, in spite of the marked decrease of the total HDL fraction.

High density lipoproteins are isolated from human plasma by ultracentrifugation in the density range 1.063–1.210 g/ml (1). The major component of HDL,[1] apoprotein AI, is the cofactor in the activation of plasma lecithin-cholesterol acyltransferase (EC 2.3.1.43) (2); elevations or reductions in plasma HDL associated cholesterol levels are inversely related to the relative risk of atherosclerotic vascular disease (3, 4).

Studies on the analytical ultracentrifugation pattern of HDL showed a bimodal distribution, interpreted in terms of two HDL subfractions; HDL_2 (d 1.063–1.125 g/ml) and HDL_3 (d 1.125–1.210 g/ml) (5). This bimodal distribution was confirmed by rate flotation in zonal rotors (6); however, more recently, using density gradient centrifugation and gradient gel electrophoresis, Anderson *et al.* (7) described at least three homogeneous major components in the HDL particle size distribution; these authors also developed a method for quantitatively resolving the total HDL schlieren pattern in HDL subcomponents (8). By contrast, Kostner and Laggner (9), using steric exclusion chromatography, concluded that HDL is "a polydisperse system of macromolecules with a unimodal particle size distribution." Finally, subfractionation of HDL by heparin-Sepharose affinity chromatography yields three fractions, one of which contains apoprotein E and binds to cell surface receptors (10).

Despite the heterogeneity, indicated by most authors, HDL particles contain, in general, both apoprotein AI (apo-AI) and apoprotein AII (apo-AII). The molar ratio between these apoproteins may vary widely among the different subfractions (6, 11), but the importance of apo-AI and apo-AII in maintaining the structure of the HDL particle has been clearly established (12, 13). In spite of the fact that both primary and secondary structures of these apoproteins have been determined, some controversies exist about their higher orders of conformation (14, 15). Recently, the quaternary structure of apo-AI and apo-AII has been investigated by chemical cross-linking with dimethylsuberimidate (16, 17). In this system, apo-AI self-associates to give tetrameric and pentameric species; in contrast, apo-AII gives only dimeric forms. The same bifunctional reagent was employed for the analysis of the quaternary structure of HDL from different animal species (18). Cross-linked human HDL_2 and HDL_3 gave diffuse bands on SDS gels, centered at 120,000 and 90,000 daltons, respectively, in good agreement to the expected amount of protein in the lipoproteins, as calculated from compositional studies (19, 20). Cross-linking with DMS appears to be a useful tool for studying the quaternary structure of lipoproteins.

The detection of an apoprotein AI variant, defined as apo-AI_{Milano}, in the HDL fraction of five members of the same family, was recently described by our group (21–23). The abnormal apo-AI, characterized by cysteine content (22), forms complexes with apo-AII (AI_M-AII) as well as dimers. The HDL particles in these subjects present morphological and biochemical abnormalities, being relatively enriched with triglycerides and containing less cholesterol esters than normal HDL (23). The protein content of the particles is in the normal range, but protein composition is obviously altered, due to the presence of two large protein constituents of molecular weight 35,000 (AI_M-AII) and 55,000 (AI_M-AI_M), besides apo-AI and apo-AII (22).

In the present study, the composition of the HDL_3 subpopulations, isolated by preparative ultracentrifugation and agarose-gel filtration, was examined. Their quaternary structure, as assessed by the use of the cross-linking reagent DMS, was investigated, in order to elucidate the role of the AI_M variant in determining the final structure of HDL particles.

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[1] The abbreviations used are: HDL, high density lipoproteins; AI_M, AI_{Milano}; VLDL, very low density lipoproteins; DMS, dimethylsuberimidate; IEF, isoelectric focusing; SDS, sodium dodecyl sulfate; CL, cross-linked; PL, phospholipids; TC, total cholesterol; TG, triglycerides; EC, esterified cholesterol; FC, free cholesterol.

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EXPERIMENTAL PROCEDURES

Subjects—Studies were carried out in apo-AI$_M$ affected subjects numbered I-1, II-2, and II-3 in the previously published pedigree (23). Individual or pooled plasma from these subjects was used. Age-matched normolipidemic subjects acted as controls.

Lipoprotein Isolation—Plasma containing Na_2-EDTA (1 mg/ml) was obtained from fasted subjects. Plasma density was raised to d = 1.063 using solid KBr and the $d > 1.063$ lipoprotein fraction was isolated by centrifuging for 20 h at 45,000 rpm in a 50Ti rotor (Beckman, Palo Alto, CA). HDL were separated at the density of 1.21 for 40 h at 45,000 rpm; in some experiments, the top fraction was dialyzed against a KBr-NaCl-EDTA solution of d = 1.125 and centrifuged at 45,000 rpm for 40 h to isolate HDL$_2$. Total HDL, HDL$_2$, and HDL$_3$ were washed and concentrated by centrifuging at their upper density using an SW 41Ti rotor, and then extensively dialyzed against 0.15 M NaCl, 10^{-3} M EDTA, pH 7.4, at 4 °C.

Agarose Gel Chromatography—The gel chromatographic separation of lipoprotein fractions after ultracentrifugation was performed on a Bio-Gel A-5m column (Bio-Rad) (1.5 × 90 cm). The same amounts of lipoprotein-protein were applied to the column and the column was eluted with 0.01 M Tris-HCl, pH 7.5, 0.15 M NaCl, 0.01% EDTA, and 0.01% NaN_3 at 4 °C, fractions of 1.7 ml being collected. The column was calibrated with ferritin (Stokes radius 59 Å), catalase (52 Å), and aldolase (46 Å).

Chemical Cross-linking—Cross-linking experiments were carried out according to published methods (24) using dimethylsuberimidate (Sigma) as bifunctional reagent. Briefly, DMS solution (25 mg/ml in 2 M triethanolamine-HCl, pH 9.0) was added to lipoprotein or protein solutions and the reaction carried out at room temperature for 180 min. The sample was then added with Tris/glycine buffer (pH 8.3), SDS, mercaptoethanol (final concentrations 2.5 mM, 3% and 0.02%), and incubated at 37 °C for 90 min before the application to SDS gels.

SDS Gel Electrophoresis—Lipoprotein fractions were delipidated with chloroform/methanol (2/1) after dialysis against 0.15 M NaCl, 10^{-3} M EDTA, pH 7.4. SDS-polyacrylamide gel electrophoresis was carried out according to Stephens (25) using 5% acrylamide gels for cross-linked samples and 10% acrylamide gels for delipidated samples. After staining in standard Coomassie blue-methanol-acetic acid solution, protein bands were quantitated by densitometry (Seroskope 160, Elvi, Logos, Milan, Italy). The optical density recorded for each group of bands increased linearly with increasing amounts of loaded proteins. For the molecular weight estimations of cross-linked lipoproteins, proteins similarly cross-linked with DMS were used as standards: apoprotein AI (M_r = 27,500–137,500), aldolase (M_r = 39,500–158,000), bovine serum albumin (M_r = 67,000–134,000). Molecular weights of unknown samples were calculated from the plot of relative mobility against log molecular weight.

Isoelectric Focusing—Polyacrylamide gel isoelectric focusing was carried out in rod gels (0.5 × 9 cm) as previously described (21). After staining, protein bands were quantitated by densitometry.

Other Methods—Total and free cholesterol (26), triglyceride (27), and phospholipid (28) determinations were carried out by enzyme methodologies. Protein analysis was by the method of Lowry *et al.* (29). Distribution of lipoproteins after gel chromatography was monitored by recording the absorbance at 280 nm of the fractions.

RESULTS

Total HDL from patients with the apo-AI$_M$ variant and from healthy controls were similarly isolated by preparative ultracentrifugation. The pattern of HDL fractionation did not differ in the three examined subjects with the AI$_M$ variant. Samples were therefore pooled for column separations, whereas separate gels were obtained from each subject in the case of whole lipoprotein fractions.

Cross-linking studies on total HDL demonstrated a bimodal distribution in this fraction, when isolated from control subjects; in SDS gels, cross-linked HDL show two distinct broad bands centered at M_r = 120,000 and 90,000 (Figs. 1–3). These bands, apparently identical with those shown by Swaney (18), were identified as HDL$_2$ and HDL$_3$, respectively. When total HDL from AI$_M$ subjects were similarly cross-linked with DMS, the pattern observed in SDS gels was markedly different from that of control CL-HDL, with the appearance of two bands centered at a molecular weight of 70,000 and 58,000 (Fig. 2).

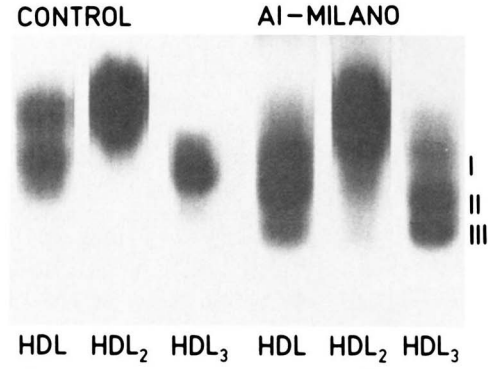


FIG. 1. **SDS gel electrophoretic separations of cross-linked lipoproteins from control and AI$_{Milano}$ subjects.** Purified lipoproteins (30 μg of lipoprotein-protein) were cross-linked with DMS and applied to 5% acrylamide SDS gels. A similar mobility is shown for *HDL$_2$* particles (highly concentrated in the case of the AI$_M$ samples), whereas three different subpopulations of HDL$_3$ (*I, II, III*) are noted in the AI$_M$ subjects.

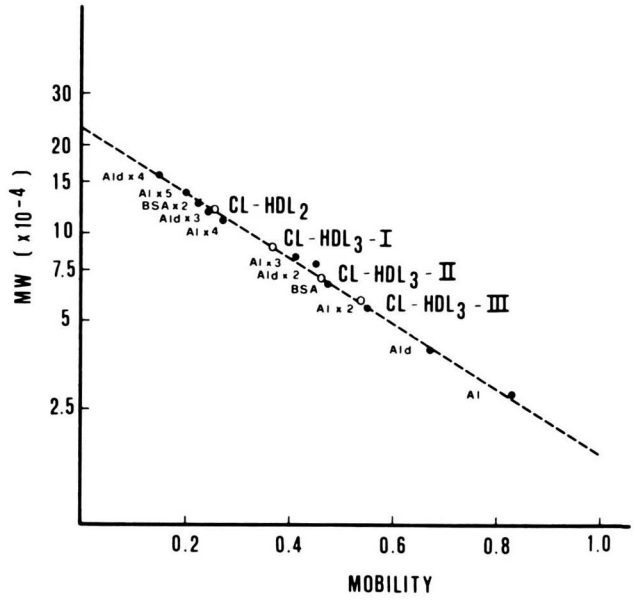


FIG. 2. **Relative mobility on SDS gels of different protein standards and of cross-linked HDL$_2$ and HDL$_3$ from AI$_M$ subjects.** Protein standards and lipoproteins (30 μg of protein) were cross-linked with DMS and applied to 5% acrylamide SDS gels. Three different subpopulations of *HDL$_3$* (*I, II, III*) are separated, with apparent M_r = 90,000, 70,000, and 58,000, respectively.

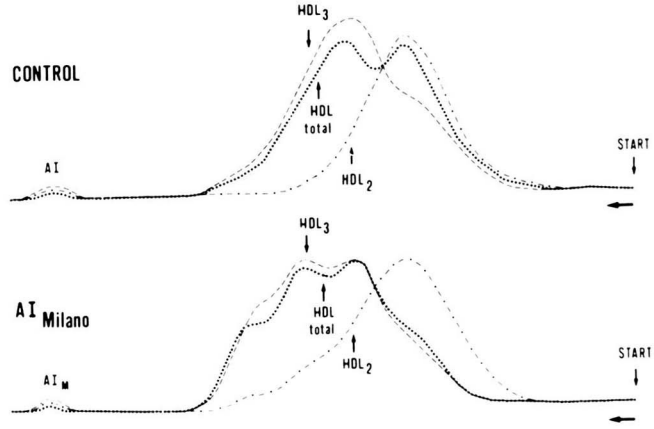


FIG. 3. **Densitometric scans of the SDS gels shown in Fig. 1.** The expected bimodal distribution of total HDL is shown in the control sample. In the *AI$_M$* lipoproteins, four different components are detected in total HDL. Three of these may be separated in the HDL$_3$ fraction.

The subfractions HDL_2 and HDL_3 were isolated by preparative ultracentrifugation and then characterized by cross-linking with DMS. CL-HDL_2 from both patients and controls showed a nearly identical pattern (Figs. 1 and 3) with a large band at M_r = 120,000. By contrast, CL-HDL_3 from AI_M subjects showed a marked heterogeneity, as compared to control CL-HDL_3. The band at M_r = 90,000 was reduced and the two bands at the lower molecular weight, detected in total CL-HDL, became apparent. The three bands observed in CL-HDL_3 fractions from the AI_M subjects were identified as: HDL_3-I (M_r = 90,000), HDL_3-II (M_r = 70,000) and HDL_3-III (M_r = 58,000) (Figs. 1 and 2).

In view of the apparently normal CL-HDL_2 pattern in the patients and of the observed alterations in CL-HDL_3, further studies were devoted to chemical and structural analyses of this latter HDL subfraction. HDL_3 from patients and controls were similarly filtered through a calibrated column of 6% agarose gel; the chemical composition of the eluate was monitored by the determination of protein, phospholipid, free and esterified cholesterol, and triglycerides (Fig. 4). Both HDL_3 from controls and patients eluted as a symmetrical peak at a position intermediate between ferritin and catalase; the corresponding Stokes radius was calculated as 51.3 ± 4.1 Å.

The elution profile of HDL_3 from AI_M subjects showed a larger peak (possibly indicative of an increased particle heterogeneity) and a markedly different distribution of individual components as compared to HDL_3 from controls. PL and TG content of all fractions was increased and total cholesterol content decreased in HDL_3 from patients (Fig. 4).

The elution peak of HDL_3 was arbitrarily subdivided into six successive lipoprotein subfractions (Fig. 4), each being examined by chemical analysis, apoprotein distribution, and by cross-linking. The chemical composition of the subfractions is reported in Table I. Lipid abnormalities in HDL_3 from the AI_M subjects involve constituents of both lipoprotein core and surface; however, calculated content and volume of core (TG + EC) and surface (protein + PL + FC) constituents per lipoprotein particle were nearly identical, both in patients and controls. HDL_3 subfractions from patients constantly showed increased TG contents with a corresponding decrease of EC; FC was decreased and PL increased, as compared to control subfractions. Protein content increased progressively with decreasing particle size both in patients and in controls, while PL decreased constantly in controls and irregularly in patients.

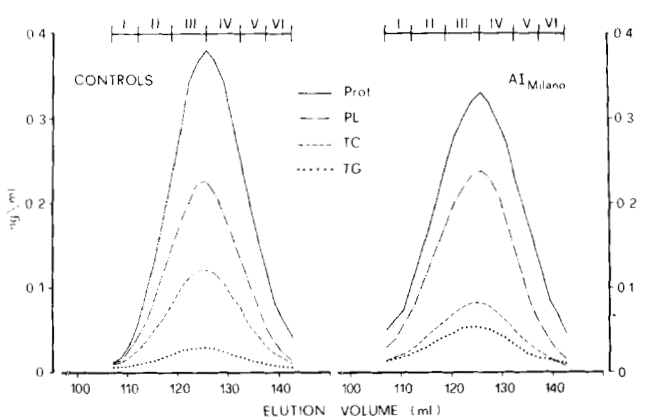


FIG. 4. **Elution pattern of control and AI_M HDL_3 from a 6% agarose column.** Ultracentrifugally purified HDL_3 from *AI_M* subjects and from *controls* were filtered through a Bio Gel A-5 m column (1.5 × 90 cm). A similar amount of lipoprotein-protein was applied to each column and eluted with 0.01 M Tris-HCl, pH 7.5, 0.15 M NaCl, 0.01% EDTA, and 0.01% NaN_3, fractions of 1.7 ml being collected. A relative enrichment of TG is noted in the AI_M lipoproteins. The six different *subfractions*, later analyzed by chemical composition and by apoprotein distribution, are indicated.

tients.

The apoprotein composition of subfractions from the agarose column was examined by SDS-gel electrophoresis. The extinction coefficients of the different protein bands, obtained by densitometry as previously described (23), were used to calculate molar ratios for the apoproteins. In controls, a constant decrease of the apo-AI/apo-AII ratio was observed, with decreasing particle diameter; AI/AII molar ratios varied from 5.5 ± 0.3 for fraction I to 2.1 ± 0.4 for fractions IV and V, again increasing to 2.9 ± 0.7 in fraction VI. In the patients, the pattern was much more complex, due to the presence in lipoprotein subfractions of multiple protein bands (AI_M, AII, dimers, and complexes). Table II reports the percentage of distribution of the individual proteins in agarose subfractions from the AI_M subjects. The content of AI_M monomer increased progressively with decreasing particle diameter, whereas that of the larger AI_M dimers decreased. The content of the AI_M-AII complex increased with a maximum at fraction IV, decreasing thereafter. The apo-AII content, generally minimal, was variable.

Apoprotein distribution of the agarose subfractions was also examined by analytical isoelectric focusing. The total IEF pattern was subdivided by densitometry of the stained gels into seven regions (Fig. 5). The nature of the corresponding protein bands is not always clear. As previously described (23), band 1 corresponds to isoprotein AI_1, the major isoprotein, with bands 2 and 4, of the monomer AI_M. The complex

TABLE I

Chemical composition of HDL_3 subfractions eluted from a 6% agarose column

Molecular weight estimates used for calculations: AA (amino acid) = 100; PL (phospholipid) = 775; EC (esterified cholesterol) = 650; FC (free cholesterol) = 387; TG (triglycerides) = 850.

Fraction	r (A)	M_r × 10^{-5}	Protein	PL	TG	TC	FC	EC	Molecules/particle				
									AA	PL	EC	FC	TG
			w%										
I													
Cont	58.5	4.06	39.0	30.6	9.6	20.8	7.1	13.7	1583	160	86	74	46
AI_M	58.7	4.14	45.5	31.2	10.9	12.4	1.6	10.9	1884	167	69	17	53
II													
Cont	55.4	3.19	47.3	30.4	4.9	17.4	4.5	12.6	1509	125	62	37	18
AI_M	55.5	3.21	46.0	33.5	8.5	12.1	2.7	9.3	1477	139	46	22	32
III													
Cont	52.4	2.50	49.5	30.1	4.0	16.4	3.8	12.5	1238	97	48	25	12
AI_M	52.5	2.51	46.5	33.8	7.9	11.8	3.1	8.7	1167	109	34	20	23
IV													
Cont	49.1	1.96	52.0	28.6	3.5	15.9	3.7	12.2	1019	72	37	19	8
AI_M	49.3	1.96	48.3	33.3	7.2	11.2	2.9	8.2	947	84	25	15	17
V													
Cont	46.5	1.56	55.0	26.4	4.0	14.6	4.0	10.6	858	53	25	16	7
AI_M	47.5	1.56	54.7	27.4	2.4	10.7	2.4	8.3	853	55	20	10	13
VI													
Cont	44.2	1.30	57.8	21.5	5.7	15.0	5.0	9.6	751	36	19	17	9
AI_M	44.2	1.29	56.9	22.0	10.5	10.5	1.1	9.4	734	37	19	4	16

TABLE II

Relative percentages of the major HDL apoproteins in the six agarose gel fractions

X ± S.D. of the densitometric percentages taken from SDS gels of apoproteins from three replicates of the column fractionations (n = 9).

Fraction	AI_M-AI_M	AI_M-AII	AI_M	AII
I	44.2 ± 2.0	32.3 ± 1.1	22.5 ± 2.0	1.2 ± 0.3
II	35.1 ± 1.2	37.0 ± 3.0	24.6 ± 3.3	3.3 ± 0.9
III	25.8 ± 3.1	43.9 ± 2.9	26.3 ± 1.7	4.0 ± 1.4
IV	19.1 ± 2.0	49.4 ± 1.5	27.3 ± 2.5	4.2 ± 0.5
V	16.5 ± 1.8	48.6 ± 2.1	32.5 ± 0.9	2.4 ± 0.5
VI	11.1 ± 0.9	44.1 ± 0.5	43.8 ± 1.2	1.0 ± 0.1

AI_M-AII focuses in a more acidic region (23) and corresponds to bands 5, 6, and 7. The isoprotein bands corresponding to the dimer are not clearly distinguishable. By analyzing the densitometric scans from IEF gels of the different agarose subfractions, it appeared that the content of bands 1 and 2 increased with decreasing diameter of the particles (from 9.6 and 10.6 to 17.3 and 17.5%, respectively). In mercaptoethanol reduced gels, similar differences could be detected in the distribution of the five AI_M isoproteins (23).

Lipoproteins in the agarose gel fractions were cross-linked with DMS. The complex pattern shown by total CL-HDL_3 in the AI_M subjects, characterized by the presence of the three bands HDL_3-I, HDL_3-II, and HDL_3-III (Fig. 6), was evident in all the six fractions, but the relative distribution of these bands varied widely (Table III). With decreasing particle diameter, the content of HDL_3-I decreased and that of HDL_3-III increased; the content of HDL_3-II increased with a maximum at fraction III, decreasing thereafter. Some unreacted AI_M monomer is present in all the cross-linked gels and its content increases with decreasing particle diameter.

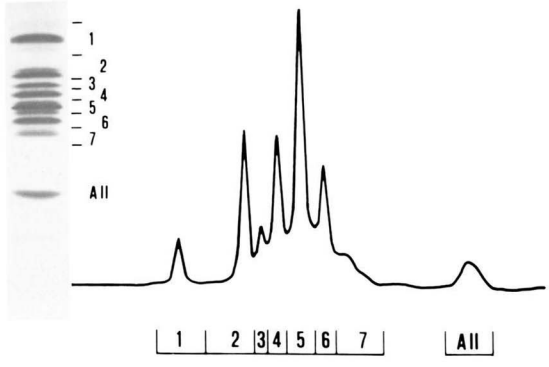


FIG. 5. **Isoelectric focusing pattern and densitometric scan from total HDL apoproteins of an AI_M patient.** Seven different *isoprotein regions* may be separated in the scan, corresponding to single or multiple isoproteins. All the lipoproteins in the six agarose fractions show a similar isoprotein pattern on IEF gels. The distribution of different isoproteins varies along the elution profile, the last of the six fractions (see Fig. 4) being enriched in isoprotein AI_1 (*region*

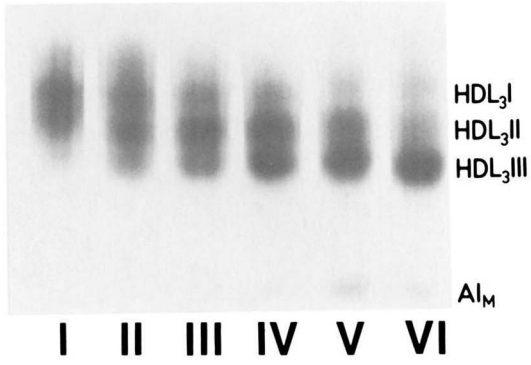


FIG. 6. **SDS gel electrophoretic pattern of the HDL_3 subfractions isolated from the 6% agarose column.** Lipoproteins in the six agarose gel *fractions* were cross-linked with DMS and applied to 5% acrylamide SDS gels. An enrichment of HDL_3-I particles is noted in the *first fraction*, whereas the *last fraction* consists almost only of HDL_3-III particles. Unreacted AI_M monomers are abundant in the *sixth fraction*.

DISCUSSION

Cross-linking studies of human HDL offer the opportunity of analyzing the quaternary structure and molecular composition of lipoprotein subspecies. DMS was selected in these studies, having a span of 12 Å, thus allowing maximal cross-linking (24); sufficiently low protein concentrations were used, so as not to allow a significant cross-linking between HDL particles (16, 24). Lipoproteins from the AI_M subjects, showing marked abnormalities in apoprotein composition, characterized by the presence of large protein complexes, may provide a useful spontaneous tool for the understanding of normal and abnormal HDL conformation.

In the AI_M subjects, differently from controls, CL-HDL_2 appear to be markedly reduced in SDS gels, while showing a cross-linking pattern not at marked variance from controls, when isolated in pure form and concentrated. It may be noted that even control subjects display an extremely wide distribution of HDL subclasses in SDS gels, after cross-linking of total HDL, in line with previously described findings with different techniques (8, 11).

HDL from the AI_M subjects are characterized by the predominance of HDL_3, separable into at least three subfractions. Agarose column fractionation of pure HDL_3 was consistent with a "polydisperse system" with a unimodal size distribution (9). However, the separation of lipoprotein particles by column chromatography, according to size, indicated a significant heterogeneity in lipid and protein components. HDL_3 from the AI_M subjects were TG enriched, with a low cholesterol and particularly EC content. In spite of these multiple abnormalities, HDL_3 fractions from patients had a core/surface ratio nearly identical with controls.

When HDL_3 from the patients, eluted from agarose column, were analyzed by DMS cross-linking, at least three particle subpopulations could be detected. The larger particles (HDL_3-I) accumulated at a relatively lower volume, the tail of the elution peak being enriched in HDL_3-III. From the apoprotein compositional data and from the molecular weight findings in SDS electrophoresis after DMS cross-linking of agarose column fractions, HDL_3-III seem to contain only the monomeric form of apo-AI; high amounts of the AI_M-AII complex and of AI_M dimers may be found in the larger HDL particles, respectively HDL_3-II and HDL_3-I.

The existence of compositional anomalies of HDL was earlier suggested, based on the *in vitro* transfer of TG from VLDL to HDL (30). More recently, the same authors showed small HDL particles (r = 44 Å) in normal humans, by using gradient gel electrophoresis (31). The small "HDL_2-like" particles in Tangier patients obviously differ from the present condition (32). The content of small, TG-enriched, so called "HDL_4" particles, was reportedly increased in hypertriglycer-

TABLE III

Relative percentages of the three HDL_3 subfractions and of free AI_M in the six agarose gel fractions

$X \pm$ S.D. of the densitometric percentages from three gels for each column fraction, collected during three different experiments (n = 9).

	Agarose-gel fractions					
	I	II	III	IV	V	VI
HDL_3-I	56.9 ± 3.3	48.1 ± 0.9	33.1 ± 3.7	17.7 ± 1.4	12.1 ± 0.1	11.2 ± 0.8
HDL_3-II	26.9 ± 3.3	33.1 ± 0.6	45.6 ± 1.0	41.7 ± 1.8	25.2 ± 2.1	17.5 ± 1.0
HDL_3-III	13.4 ± 0.8	15.5 ± 0.4	17.3 ± 1.8	36.1 ± 2.8	52.9 ± 2.0	59.7 ± 0.6
AI_M	2.8 ± 0.5	3.5 ± 0.1	4.0 ± 0.8	4.5 ± 0.4	9.9 ± 0.1	11.7 ± 1.2

idemic patients with substantially elevated serum concentrations of VLDL (31). HDL_4 may be an intermediate product in the HDL-VLDL cholesterol transfer chain (33). Small HDL particles may, on the other hand, also be generated during interconversion of HDL (34).

The presence of different HDL_3 subcomponents in a familial abnormality, characterized by the apparent absence of cardiovascular disease (21), suggests that particles in the HDL_3 density range may have a functional role in tissue cholesterol homeostasis. The unusual changes in the HDL_3 composition of these subjects may be of further help in explaining their protection from tissue cholesterol deposition, in spite of the marked decrease of the total HDL fraction.

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